PolySim OS

The Solver Catalog

157 interactive simulations with real numerics — every one runs in your browser, no install, no cloud.

The Wolfram Alpha of interactive simulation · polysimos.com

Flagship Engines, Fields & GPU Compute (19)

Visual Node Graph — Wire blocks into a live dataflow — the real editor. Symbolic calculus built in.

2D Fluid (CFD) — Interactive incompressible Navier–Stokes via Stam's stable-fluids method.

Heat & Wave Fields — 2D heat diffusion and the 1D wave equation, solved live.

AI Surrogate — Train an ML surrogate on our solver for instant predictions. The PhysicsX play.

GPU Compute (WebGPU) — Hundreds of thousands of particles on your GPU via WGSL shaders.

WebGPU Fluid — High-res GPU smoke advection you can stir.

GPU N-Body — Thousands of bodies, full $O(n^2)$ gravity on the GPU.

GPU PDE Solver — Steady heat/Poisson via GPU Jacobi at high res.

GPU 3D Fluid — 3D smoke, simulated + raymarched on the GPU.

Particle-Mesh N-Body — 100k+ bodies via GPU particle-mesh gravity.

3D CFD — 3D Navier–Stokes plume with z-slices.

3D N-Body — Gravitation in 3D with a drag-to-orbit camera.

3D FEA Space Frame — 3D structural analysis; orbit the deformed tower.

3D Heat Diffusion — Volumetric heat with slices and orbit view.

FEA Truss — Finite-element structural analysis: forces & deflection.

Meshing + BCs — Paint boundary conditions; solve steady heat.

Optimize + UQ — Gradient descent and Monte-Carlo uncertainty.

RLC Circuit — Step response and damping regimes.

Gradient Descent — Optimization on loss landscapes.

Physics, Mechanics & Thermodynamics (25)

Particle / N-Body — Gravity, orbits, and impulse collisions with a real symplectic integrator.

Electrostatics — Charges, potential heatmaps, and field lines.

Double Pendulum — The textbook chaotic system, integrated with RK4.

Projectile Motion — Ballistics with air drag; tune angle and speed.

Ising Model — A live magnetic phase transition via Monte Carlo.

Strange Attractors — Lorenz, Rössler, Aizawa & more in 3D.

Wave Interference — Two-source ripple-tank fringes.

Random Walk — Brownian motion and diffusion.

Kepler Orbits — Conic-section orbits under gravity.

Double-Slit — Quantum interference + diffraction.

Cloth / Spring-Mass — Verlet cloth you can grab and swing.

Gravity Well — Curved-spacetime rubber sheet + orbit.

Magnetic Pendulum — Fractal basins of attraction.

Percolation — A phase transition in connectivity.

DLA Growth — Branching fractals from random walks.

Ray Optics / Lenses — Trace rays through lenses.

Gas in a Box — Kinetic theory → pressure (PV=nRT).

Elastic Collisions — Momentum & energy conservation.

Buoyancy — Archimedes — float or sink by density.

Rocket Equation — Tsiolkovsky Δv budget calculator.

Blackbody Radiation — Planck curve + Wien's law.

Pendulum Wave — Phase art from many periods.

Orbital Transfer — Hohmann transfer Δv.

Standing Waves — Harmonics, nodes & antinodes.

Doppler Effect — Moving-source wavefronts & sonic boom.

Waves, Optics & Acoustics (6)

Snell's Law — Refraction & total internal reflection.

Diffraction Grating — N-slit interference maxima.

Beats — Two close tones interfere.

Curved Mirrors — Concave/convex ray tracing.

Prism Dispersion — White light into a spectrum.

Fourier Transform — Signal → frequency spectrum.

Mathematics (22)

Dynamical Systems — Lorenz, SIR epidemics, pendulums, predator–prey, and reaction–diffusion.

Symbolic Math — Parse, differentiate, simplify, solve, and plot — a real CAS in your browser.

Vector Fields — Plot any F(x,y) as a live quiver diagram.

Notebook — Cells of prose + symbolic math + compute.

Fractal Explorer — Zoom into the Mandelbrot and Julia sets.

Fourier Series — Build waves from sine harmonics.

Function Grapher — Plot up to three functions at once.

3D Surface Plotter — Orbit any $z = f(x, y)$ surface.

Matrix Calculator — Multiply, invert, determinant, transpose.

Cellular Automata — Wolfram rules & Conway's Life.

Taylor Series — Polynomial approximation, live.

Newton's Method — Root finding, tangent by tangent.

Distributions — Normal, Poisson, binomial & more.

Complex Functions — Domain coloring of f(z).

Lissajous Curves — Two sine waves, endless patterns.

Bifurcation Diagram — Logistic map route to chaos.

Direction Field — Slope fields + solution curves.

Riemann Sums — Integrals as stacked rectangles.

Eigenvectors — 2×2 transforms & invariant directions.

Markov Chains — Converge to a stationary distribution.

Parametric Grapher — Trace x(t), y(t) curves.

Euler vs RK4 — ODE integration accuracy.

Chemistry & Materials (8)

Molecular Dynamics — Lennard-Jones atoms; melt a lattice.
Titration Curve — pH vs base added; equivalence point.
Reaction Kinetics — Rate laws & Arrhenius decay.
Chemical Equilibrium — Le Chatelier shifts.
pH Calculator — pH/pOH and the acid-base scale.
Radioactive Decay — Half-life exponential decay.
Maxwell-Boltzmann — Molecular speed distribution.
Ideal Gas Law — PV = nRT with a live piston.

Computer Science & AI (10)

Sorting Visualizer — Watch algorithms sort in real time.
Boids Flocking — Emergent flocking from three rules.
A* Pathfinding — Shortest path, draw walls.
Maze Generator — Carve & solve perfect mazes.
Neural Network — Train an MLP, watch it learn.
k-Means Clustering — Unsupervised clustering, live.
Convex Hull — Wrap points in O(n log n).
L-System Fractals — Grow plants from grammars.
Turing Machine — The model of computation.
Image Convolution — Kernels behind every CNN.

Complex Systems & Chaos (9)

Traffic Flow — Phantom jams from random braking.
Forest Fire Model — Self-organized criticality.
Abelian Sandpile — Avalanches build a fractal.
Schelling Segregation — Mild bias, sharp segregation.
Langton's Ant — Two rules, emergent highway.
Reaction-Diffusion — Gray-Scott Turing patterns.
Elementary CA — Wolfram's 256 rules.
Genetic Algorithm — Evolve toward the peak.
Ant Colony Optimization — Swarm-solved TSP.

Biology & Medicine (10)

Epidemic Network — Agent-based SIR on a contact graph.
Spatial Predator–Prey — Agent-based ecology with waves.
SIR Epidemic — Outbreak compartment model.
Hodgkin-Huxley Neuron — Action potential spikes.
Lotka-Volterra — Predator-prey cycles.
Hardy-Weinberg — Allele frequency & selection.
Enzyme Kinetics — Michaelis-Menten curves.
Logistic Growth — Population carrying capacity.
Genetic Drift — Wright-Fisher allele drift.
Sequence Alignment — Needleman-Wunsch DNA.

Astronomy & Space (8)

Exoplanet Transit — Light-curve transit method.
H-R Diagram — The map of the stars.
Lagrange Points — Restricted 3-body equilibria.
Roche Limit — When tides shred a moon.
Hubble's Law — The expanding universe.
Telescope Optics — Resolution & magnification.
Stellar Parallax — Distance from angle shift.
Escape Velocity — Speed to leave a world.

Earth & Climate & Environment (8)

Energy Balance — Greenhouse equilibrium temp.
Daisyworld — Planetary self-regulation.
Milankovitch Cycles — Orbital forcing of ice ages.
Tsunami Propagation — Shallow-water wave shoaling.
Carbon Cycle — Reservoir box model.
Earthquake Shaking — Magnitude, energy, intensity.
Groundwater Drawdown — Well cone of depression.
Lapse Rate — Atmosphere & cloud base.

Finance & Quantitative (8)

Black-Scholes — Option pricing + Greeks.
Option Payoff — Strategy P/L diagrams.
Monte Carlo (GBM) — Price-path distributions.
Efficient Frontier — Portfolio risk vs return.
Value at Risk — VaR & expected shortfall.
Bond Pricing — Price, yield & duration.
Compound Interest — Investment growth over time.
Loan Amortization — Mortgage payment breakdown.

First Responders (Fire · Police · EMS · Hazmat) (8)

Wildfire Spread — Wind + slope fire behavior.
Hazmat Plume — Gaussian dispersion + PAD.
Building Evacuation — Egress flow & clear time.
START Triage — Mass-casualty triage tool.
Fire Hose Hydraulics — Friction loss & pump pressure.
Skid-to-Stop Speed — Accident reconstruction.
Radio Range — VHF/UHF link budget.
Blast Standoff — Overpressure & safe distance.

Structural & Civil Engineering (8)

Beam Deflection — Sag & bending moment.
Column Buckling — Euler critical load.
Mohr's Circle — 2D stress transformation.
Shear & Moment — SFD/BMD diagrams.
Retaining Wall — Rankine earth pressure.
Seismic Base Shear — Equivalent lateral force.
Soil Bearing — Terzaghi capacity.
Concrete Beam — RC flexural capacity.

Electrical & Signal Processing (8)

Bode Plot — Frequency response.

Filter Designer — RC/RLC filter response.

PID Controller — Step-response tuning.

Transmission Line — VSWR & reflections.

Op-Amp Circuits — Amplifier configurations.

Three-Phase Power — Grid power & phasors.

Transistor Bias — BJT Q-point load line.

Sampling & Aliasing — The Nyquist theorem.